

Extended Analysis

R1–R6 expand analyses referenced from the main text. R7 expands the binary KEGG sensitivity in Figure 2d and the post-hoc exploratory all-target analysis summarised in Figure 3d. Each follows the corresponding Methods section.

R1. Burden-heterogeneity model diagnostics

These quantities support the single sentence in Results that a constant per-edge reversal probability is rejected and that the dispersion is not a by-product of statistical power. They are diagnostics rather than tests: the working model treats regulators as independent, which targets shared between regulators violate, so the within-target permutation reported in the main text calibrates the outside-interval comparison and these nominal reference distributions do not.

Treating each regulator’s reversals as k_i successes out of its n_i both-significant edges, a constant per-edge reversal probability is rejected (Tarone $Z = 252$), and a beta-binomial model retains an intra-class correlation of $\rho = 0.072$ after adjusting for a positive degree slope of $+0.431$ per $\log_{10}$ out-degree (dispersion likelihood-ratio $\chi^2 = 2,853$; slope $\chi^2 = 19.8$, $p = 8.7 \times 10^{-6}$).

Differential statistical power does not explain the dispersion. Adding each regulator’s achieved precision (the median standard error of its own edges), its on-target knockdown efficacy (taken from the self edges excluded from the trans estimand) and a cubic rather than linear degree term leaves $\rho = 0.0635$ against 0.0661 for degree alone on the same 555 regulators, and measured per-cell support changes nothing further on the 138 regulators that have it ($\rho = 0.0566$ against 0.0568).

A legacy sensitivity over the 264 regulators with at least 50 both-significant trans edges, using a two-sided leave-one-out decile binomial, called 99 (41 enriched, 58 depleted) on that narrower family. The attempted degree-matched complex permutation was not uniformly calibrated, with false-positive rates from 2.0% to 12.5% among testable complexes and two small classes not testable, so we make no complex-level enrichment claim. Functional groupings are retained only as descriptive annotations of where reversals occur.

Figure 1e,f shows the whole-family concentration and degree-adjusted rate views. Degree adjustment does not remove the heterogeneity, but it does not follow that connectivity is irrelevant or that the residual variation is an intrinsic regulator property.

The separate post-hoc regulator-direction-purity sensitivity gave 12 pure regulators among 157 carrying at least four reversals (null expectation 7.110; one-sided exact $p = 0.00322$), 10/139 at a threshold of five (expectation 5.455; $p = 0.000991$), 8/131 at six (expectation 4.959; $p = 0.0106$) and 3/117 at seven (expectation 1.861; $p = 0.230$). These thresholds were inspected after the catalogue was seen, so the values are reported without a multiplicity-adjusted claim.

R2. Catalogue effect sizes and positive controls

A reversal would be trivial if one endpoint sat at zero, because its sign would then be arbitrary. The census design rules that out: an edge qualifies only when it is significant in both states, so its weaker side must still clear the source threshold, and across all 4,379 edges the smallest weaker-side absolute z-statistic is 1.98. That is a property of the selection rule, not a calibrated probability that each edge truly reverses. Significance is not size, and roughly half the catalogue is modest: the

median weaker-side absolute log2 fold change is 0.245, a 1.19-fold change (Supplementary Figure S1a). We therefore keep all 4,379 as the threshold-defined catalogue and report separately the 532 edges exceeding 0.5 in both states (1.41-fold) and the 100 exceeding 1.0 (twofold); within the 532-edge core the weaker-side absolute z-statistic has a minimum of 2.04 and a median of 2.93.

Known context-dependent regulators appear among the higher descriptive rates (Supplementary Data 2): CBFB at 16.0%, STAT5B at 15.6%, STAT3 at 14.8% and BATF at 14.8% sit roughly threefold above the 5.2% conditional baseline, and none belongs to the co-regulator classes that lead the raw counts. These rates divide by each regulator’s own both-significant edges, and the panel is a face-validity check, not a confirmatory enrichment test. One expected family is absent by construction, since no NFAT member reaches the required out-degree in both states: NFAT5 carries 1,125 significant targets at Rest and two at Stim48hr (Limitations).

Individual exemplars show large opposite-sign knockdown coefficients (Supplementary Figure S1c), displayed only for edges with adjusted $p < 0.01$ in both states, an order of magnitude stricter than the catalogue threshold. TAF6L knockdown raised IFIT2 at Rest (log2 fold change +3.23) and lowered it at Stim48hr (−3.33); CARMIL2 knockdown lowered GZMA at Rest (−2.11) and raised it at Stim48hr (+2.21). GZMA, IL13, CCR5 and CCR7 are canonical activation and effector genes. These examples establish the direction and magnitude of the measured perturbation responses, not direct action by the intact regulator.

R3. The unselected 8 h timepoint

The catalogue compares the two endpoints of the activation trajectory. The atlas also released differential expression at 8 h from the same donors, assay and pipeline, covering 323 of 324 reversal-bearing regulators and 4,377 of the 4,379 reversals. That condition selected no catalogue edge, so it is a non-selection timepoint for the same net knockdown effects, but it is not an independent cohort (Methods). Values are in `results_data/census/eight_hour_switch_timing.json`.

Whether the 8 h effect lies between the endpoints is uninformative, because an opposite-sign edge spans zero and noise centred on zero also lies between them: substituting simulated noise placed the effect 0.53 of the way along the interval against 0.46 observed. Requiring 8 h significance would instead select a trajectory and exclude effects near zero. We therefore correlated the 8 h z-statistic with the Rest z-statistic within each regulator, against a permutation breaking target identity while retaining each regulator’s 8 h distribution, and against a noise simulation.

The 8 h statistics depart from the no-effect benchmark on these edges (Supplementary Figure S1d): median absolute z 1.433 against 0.674 under the half-normal reference, with 36.1% above 1.96. Across 110 regulators the median within-regulator Rest–8 h correlation is +0.190, against a target-label permutation null near zero ($p < 0.0005$) and +0.010 under simulated noise (Supplementary Figure S1e); the 3,119 edges above an absolute z of 2.5 at both endpoints give +0.221, and stable edges give +0.719 across 561 regulators as a positive control. This is within-atlas target-specific temporal coherence, not independent validation of the endpoint reversal label.

Timing itself resolves for a minority. At an 8 h false-discovery rate below 0.10, 1,279 reversals (29.2%) are resolved and 3,098 (70.8%) are not (Supplementary Figure S1f); of the resolved, 824 retain the Rest sign and 455 match the Stim48hr sign. Retention is more frequent among lost than gained effects (529/704, 75.1%, versus 295/575, 51.3%), and across the 75 regulators carrying resolved edges of both orientations the median paired difference is 0.089 (40 positive, 14 negative and 21 tied; two-sided paired Wilcoxon $p = 0.0050$). The high-magnitude core gives 86 and 54, and

the regulator-bootstrap retaining fraction is 64.4% (95% CI 59.6–69.3%). Figure 2e shows these three columns for all 39 reversal targets of the six one-direction regulators.

Three controls bound the classification. Unresolved edges are not simply weakly measured: they carry the same Rest-to-48 h span as resolved ones (median 0.601 against 0.614, $p = 0.59$) and the same 8 h standard error. Among the 47,650 stable edges resolved at 8 h, whose endpoints share a sign and so have a known right answer, 99.2% carry it, which calibrates the estimate. A stable edge with one bad endpoint would enter the census as a reversal and read as Rest-sign-retaining, but that model predicts the worse 48 h precision for such edges and they carry the better ($p = 0.052$, opposite to the artefact).

A separate post-hoc sensitivity, retained in Supplementary Data 17 and 18 rather than used as a main panel, found no detectable excess association between independent target activation and absolute coefficient movement for reversing versus stable edges (median within-regulator $\rho = 0.077$ versus 0.055 across 154 paired regulators; two-sided paired Wilcoxon $p = 0.572$). This negative comparison is not an equivalence test and does not weaken the distinct sign-state result in Figure 2f.

The lean is descriptive and not evidence of a late-firing switch. The donor-reproduced subset leans the same way, 40 of 62 resolvable edges (64.5%), but spans twelve regulators with only four meeting the five-edge minimum for a clustered test, so it cannot establish timing independently of edge selection (Limitations). The conditional fraction varies among regulators (median 0.667, interquartile range 0.425–0.800) without tracking the number of resolvable reversals ($\rho = +0.019$, $p = 0.90$) or out-degree ($\rho = -0.051$, $p = 0.72$), and we did not show that this variation replicates across donors.

No functional class accounts for it, and we report the pre-registered negative (Supplementary Figure S1g). Chromatin-annotated regulators gave a median retaining fraction of 0.600 against 0.667 for the remainder ($p = 0.093$), while Gene Ontology annotation density tracked that fraction ($\rho = -0.448$, $p = 0.0011$, $n = 50$): the exploratory functional categories were confounded by how thoroughly genes had been studied. Timing-sign agreement was 84.0% among candidates testable across disjoint donor partitions (cluster-bootstrap 95% CI 78.1–87.8% over 61 regulators) against a within-regulator permutation null of 65.1% ($p < 10^{-4}$), an internal selected-subset reproducibility check rather than evidence for a regulator-specific mechanism.

R4. Separate-dataset recurrence: sensitivity and exemplars

Arce provides the formal separate-dataset recurrence test. The comparison is confined to the 241 reversal edges it covers, and because that library was selected for state-specific factors and comes from the same laboratory, only the within-regulator reversal-versus-stable contrast is interpretable and the percentages are not genome-wide rates (Limitations).

Catalogue reversals had smaller effects than stable edges, so the recurrence could have been an effect-size artefact. Matching on magnitude within regulator achieved balance and gave an odds ratio of 3.13 (95% CI 2.07–4.91) against 3.23 (2.32–4.46) for the unmatched regulator-stratified estimate, so it is not.

MED12 shows why a per-edge rate and an aggregate state score are different estimands. Its catalogue reversal rate is 5.3%, near the conditional baseline, yet 43.4% of its covered reversal edges reverse again in Arce against 14.5% of its stable edges (odds ratio 4.52, $p = 1.0 \times 10^{-6}$; Supple-

mentary Figure S2e). A regulator can therefore be unremarkable by rate and still carry edges that recur strongly.

CBFB->IL2RA has opposite signs in the Arce Perturb-seq data and in a CRISPR-knockout surface-protein screen (Supplementary Figure S2f). This is a recurrence across assays within one study rather than an independent biological replication, and it is the anchor for the proposed validation design (Supplementary Table S1).

Song et al. supplied a broader independent-laboratory Jurkat sensitivity with coefficients in both states for 297 catalogue reversals and 1,556 same-regulator stable edges across 71 regulators. Sign reversal occurred for 151/297 catalogue reversals (50.8%) and 695/1,556 stable edges (44.7%), giving a descriptive Mantel–Haenszel odds ratio of 1.26 and a median within-regulator difference of +0.026. The high-magnitude and precision-stable subsets showed the same small aggregate excess. Only two focal regulators had measurable self-transcripts in the targeted expression panel, below the pre-specified minimum of ten, so the planned confidence interval and inferential tests were withheld. Song is therefore a descriptive cross-system projection, not confirmatory validation.

The Southard Hs27-to-RPE-1 CRISPRa comparison [southard2025comprehensive] provided a pre-specified cross-cell-type structural test rather than T-cell replication. Of 392 exact-ID-mapped catalogue reversals, 86 formed matched sets across 11 regulators. Sign switching occurred for 43/86 reversals (50.0%) and 69/140 matched stable effects (49.3%); the regulator-balanced risk difference was -0.0687 (two-sided plus-one permutation $p = 0.735$). The primary adequacy gate passed but the support gate did not. The secondary 49-regulator family contributed only 61 matched reversals across four regulators and failed its adequacy gate (risk difference -0.145 ; $p = 0.840$).

The independently defined Southard switch network contained 66 high-confidence edges from 25 regulators to 52 targets. Its target-convergence statistic was 18 against a stratified reallocation-null mean of 1.543 and null maximum of 12 in 1,000,000 draws (upper-tail plus-one $p = 1.0 \times 10^{-6}$; Holm-adjusted $p = 2.0 \times 10^{-6}$). Eleven targets received at least two switch edges. Only two targets had in-degree at least three and neither was single-orientation, so the pre-specified adequacy gate for that second property failed. The supported result is therefore target convergence in a separate cross-context CRISPRa network, not recurrence of the T-cell edge catalogue, stimulation replication or evidence for any individual target.

R5. External promoter-island accessibility analysis

To test whether baseline or early-stimulation chromatin accessibility at target promoter regions accounts for the direction in which a target’s reversals run, we analyzed external promoter-island ATAC-seq data from primary naive CD4+ T cells before and 5 h post-stimulation with anti-CD3/CD28 (GSE116696; Yukawa et al., 2020). All 903 specificity targets mapped to the promoter dataset, and 231 targets carried non-zero promoter opening scores at 5 h.

The target signed reversal score correlated weakly with early promoter opening (Spearman $\rho = 0.0284$), remaining below the target-conditioned null mean ($\rho = 0.0419$, 95% null interval -0.0050 to 0.0882 , plus-one $p = 0.716$) and the regulator-margin null mean ($\rho = 0.0321$, $p = 0.558$). Specific contrast hypotheses also failed significance: the CD28-dependent contrast ran opposite to prediction ($\rho = -0.0630$, Holm-corrected $p = 0.973$), and the nominal A-FOS contrast ($p = 0.0296$ under the target null) failed its predeclared information gate (59/903 informative targets), Holm correction ($p = 0.0591$), and the regulator-margin sensitivity ($p = 0.0951$).

These processed promoter-island data indicate that early promoter accessibility does not account for target reversal orientation beyond chance expectation. Because this dataset measures processed promoter-island peaks in unperturbed naive T cells rather than matched perturbation-accessibility profiles across activation, this negative result does not rule out distal enhancer dynamics, locus-specific chromatin remodelling, or context-dependent co-regulator complex recruitment.

R6. Target function and reversal orientation, matched on target induction

Among the 279 single-orientation targets of Figure 3b, 161 carry only lost effects and 118 only gained effects. We asked whether that orientation carries functional information, using each group as the other’s background so that being a reversal target cancels rather than being re-detected against the genome.

An unmatched comparison is not interpretable here. A target’s own median expression change between Rest and 48 h differs sharply by orientation: $+1.253 \log_2$ for lost-group targets against -0.044 for gained-group targets (Mann-Whitney $p = 1.5e - 27$). Sorted into induction quartiles, the lowest quartile splits 9 lost to 61 gained and the highest 68 to 2. Any functional term whose members are unusually induced will therefore appear orientation-associated for that reason alone.

We repeated the comparison with orientation permuted only within induction quartile (4 bins, 10,000 draws, seed 20260903), across two families fixed for this analysis but not preregistered: the 7 testable sets from a 19-term KEGG family selected and grouped after inspection of the census, and the 164 testable GO Biological Process terms, each Benjamini-Hochberg corrected within family. No term in either family reached $q < 0.10$. The cytokine association did not survive the matching: KEGG cytokine-cytokine receptor interaction moved from $p = 0.048$ unmatched to $p = 0.42$ matched, providing no evidence for a functional signal beyond the target-induction difference (the separately assembled post-hoc 12-gene cytokine set carries a median shift of $+1.403 \log_2$ against $+0.740$ for all 279).

The 12 GO terms with the smallest matched nominal p values included several cell-cycle terms; the strongest, Mitotic Sister Chromatid Segregation (GO:0000070), had matched $p = 0.0066$ but $q = 0.82$. Two sets assembled after inspecting the results are retained only as exploratory summaries: all 16 proliferation-set targets were gained-oriented (matched nominal $p = 0.0002$), while all 12 cytokine-set targets were lost-oriented (matched nominal $p = 0.033$). These values are uncorrected for the data-driven selection, and no q value is assigned. In a signal-free run with orientation shuffled before analysis, neither set was nominally significant and no declared-family term reached $q < 0.10$.

The separate Arce Perturb-seq arm contained 19 gained-oriented effects among 25 cell-cycle edges, compared with 417 among 806 other edges; 7 of 8 covered catalogue reversals were gained-oriented. These counts are descriptive because multiple edges share regulators and the dataset contains only two donors. The bulk arm contained 11 cell-cycle edges and was too small to resolve the contrast. The Song Jurkat screen was not used because it lacks a quiescent baseline.

Cell-cycle position was measured with a transcript score and with surface CD71 in the Arce protein data, but both remained coupled to mRNA sequencing depth. Two predictions of a simple composition explanation were then tested directly and neither was supported: cell-cycle reversals were not more detection-driven than other reversals (Cliff’s $\delta = -0.038$; Holm-adjusted $p = 0.70$), and their mean within-regulator coherence exceeded same-regulator, size-matched non-cell-cycle targets by only $+0.0050$, below the 80% minimum detectable effect of 0.0158 across 248 regulators. The

corrected conclusion is therefore negative: no declared KEGG or GO term predicted lost versus gained reversal orientation after matching target induction, and the post-hoc proliferation pattern remains hypothesis-generating rather than mechanistic evidence.

R7. Broad functional concentration of network-adjusted reversal burden

An initial KEGG analysis compared the 2,577 targets with at least one reversal with all 9,554 eligible targets. Conventional Fisher tests, followed by Benjamini–Hochberg correction across 113 filtered KEGG Legacy pathways, flagged 14 pathways at $q < 0.10$ before unequal reversal opportunities were considered. Targets connected to more eligible regulators had more chances to receive a reversal, so we repeated the analysis with 20,000 permutations that kept each regulator’s reversal count and eligible target pool fixed. The opportunity-aware family-wide test was negative (global $p = 0.184$; 0/113 terms passed single-step maxT correction). A separate exact in-degree sensitivity flagged p53 signalling, cell cycle and proteasome at $q < 0.10$, but none also passed the regulator-preserving maxT test. Thus, the pathway question was useful, but the ordinary binary test was not adequate for inference; we retain it as a transparent sensitivity and use the network-adjusted all-target ranking below for biological interpretation. Six of the 14 conventional candidates also appeared among the nine KEGG results from the ranked analysis below, but this is not independent replication because both analyses use the same census.

Figure 2d displays those 14 conventional candidates together with a separate signed-orientation analysis. Antigen processing and presentation contained 12 lost and 43 gained reversal edges and was the only row whose gained orientation passed both the target-conditioned maxT test ($p = 0.00344983$) and the regulator-margin maxT sensitivity ($p = 0.0000499975$). The two tests preserve complementary single margins; they are not combined and do not constitute a joint-conditioned test. This result supports gained-orientation over-representation within that annotation, not pathway activation, suppression or mechanism.

We next retained the full reversal-burden information and asked a broader question than the 279-target orientation analysis: across all 9,554 census targets, do genes in a KEGG or GO process tend to rank toward unusually high reversal burden after their network opportunities are taken into account? The ranking includes every eligible target, including 6,977 with no observed reversal, and the permutations preserve the reversal count and eligible target pool within each regulator-by-Rest-sign stratum.

Nine of 168 KEGG Legacy pathways met the repeatable within-library FDR rule in the pooled permutations and in both independent banks: cytokine–cytokine receptor interaction, prion diseases, thyroid cancer, antigen processing and presentation, proteasome, chronic myeloid leukemia, JAK–STAT signalling, apoptosis and p53 signalling. None survived the stricter global maxT correction across all 2,593 tested KEGG and GO terms. The KEGG labels also shared leading-edge genes: 31 targets occurred in at least two of the nine leading edges, and Supplementary Figure S4c displays the first 28 under the fixed order. These pathway names are therefore overlapping descriptions, not nine independent biological discoveries.

Among 2,425 GO Biological Process terms, 161 met the same repeatable within-library FDR rule. Twelve passed pooled global maxT correction, and ten remained supported in each permutation bank separately: mitotic cytokinesis; apoptotic process; organelle organisation; regulation of type II interferon production; positive regulation of apoptotic process; proteasome-mediated ubiquitin-dependent protein catabolic process; intrinsic apoptotic signalling in response to endoplasmic-reticulum stress; cellular response to lipopolysaccharide; cellular response to cytokine stimulus;

and intrinsic apoptotic signalling. Positive regulation of protein kinase activity and regulation of the ERK1/2 cascade passed pooled maxT but not the stricter both-bank stability rule. The ten stable terms point to recurring apoptosis, cytokine-response, proteasomal-turnover, cell-division and organelle-organisation annotations, while their overlapping memberships prevent them from being counted as ten independent mechanisms.

Cross-bank null calibration gave individual $p < 0.05$ rates of 0.0498–0.0502. The chance that any term passed within-family $q < 0.10$ in a signal-free draw was 0.0821–0.0828 for KEGG and 0.0096–0.0103 for GO, below the predeclared 0.12 reference. A separate full replay reproduced every scientific flag. Supplementary Figure S4a shows all nine eligible KEGG pathways, panel b shows the first 12 of 92 deterministic GO overlap representatives, covering 30 of the 161 eligible GO terms, and panel c shows the KEGG leading-edge overlap. The other 80 GO representatives, covering 131 eligible terms, are not plotted. Supplementary Data 19 retains the complete 2,593-term table, including all 161 statement-eligible GO terms, while the complete representative assignment remains in the figure’s source bundle.

The supported result is a rank shift toward unusually high network-adjusted pooled reversal burden among genes in these terms. It does not identify whether their reversals were lost or gained, measure pathway activity, establish a mechanism, validate an individual edge or test enrichment among the selected 279 targets in Figure 3c.